

LINUX LABORATORY

Part 1.

Case Scenario

As the Linux Administrator for fast-growing company, you have been tasked with creating, modifying, and removing user accounts from the Linux server. The company has just hired 9 new employees to fill 3 newly designed departments. The departments that have been created are Engineering, Sales and HR. The server must be setup with the appropriate files, folders, users, groups and permissions to ensure a successful launch of the newly designed departments.

Objectives

1. Create a directory at the root (/) of the file system for each department. This name should reflect the department name that will use the directory.
2. Create a group for each department. This name should reflect the department name that the group will be assigned.
3. Create an administrative user for each of the departments.
 - a. The user will have a Bash login shell
 - b. The user will belong to the respective group for each department. This will need to be the user's primary group.
4. Create two additional users for each department.
 - a. The users will have a Bash login shell.
 - b. The users will belong to their respective group for each department. This will need to be the user's primary group.
5. For security reasons, the following modifications will need to be made to each of the departments' respective directories:
 - a. Ensure that the owner of each of the directories is the department administrator and the group ownership is the group for each department.
 - b. The department administrator will have full access to their respective department directories.
 - c. Ensure that only the owner of a file in the department's directory can delete the file. The user will also have ownership of their respective department folders.
 - d. Normal users in each department will have full access (Read, Write and Execute) to their respective department folders.
 - e. The department folders will ONLY be accessible by users/administrators in each of the respective departments. Ensure that no one else will have permissions to the folders.
6. Create a document in each of the department directories.
 - a. The ownerships on this file will be the same as the directory it is located in.
 - b. The document should contain only one line of text that states, "This file contains confidential information for the department."
 - c. This file can be read by any user in the department but can only be modified by the department administrator. No one else has permissions to this file.

Case Scenario

In the User Management challenge lab, you were tasked with creating users and groups. Using the commands one at a time from the command line can be a tedious process and could lead to potential errors in syntax. It is your duty, as an administrator, to make the process as seamless and efficient as possible.

Objectives

1. Create a bash script to perform user management tasks as outlined below:
 - a. Create a new group. Each group must have a unique name. The script must check to ensure that no duplicate group names exist on the system. If a duplicate is found, an error needs to be reported, and the administrator must try another group name.
 - b. Create a new user. Each user must have a unique name. The script must check to ensure that no duplicate usernames exist on the system. If a duplicate is found, an error needs to be reported and the administrator must try another username. The user will have a Bash login shell and belong to the group that was created in the previous step.
 - c. Create a password for each user that is created.
 - d. Ensure that the new user created is a member of the new group created.
 - e. Create a directory at the root (/) of the file system with same name as the user created.
 - f. Set the ownership of the directory to the user and group created.
 - g. Set the permissions of the directory to full control for the owner and full control for the group created.
 - h. Set the permission to ensure that only the owner of a file can delete it from the directory.
 - i. Ensure that the script is executable.

This script should be designed to accept any username and any group name. **DO NOT hardcode commands to create specific usernames and group names.**

Hints

- Logical order is important
- There is a special wildcard that can be used to determine if the previous command is successful or not. This will be useful for this script.

LABORATORY ENVIRONMENT

You can choose to carry out the laboratory work on your own optional Linux distribution, or in the Linux simulator included in the course room on NETACAD.

DELIVERABLES

NOTE! This lab is a group work lab meaning ALL members of the group should contribute equally to all work.

All group members must be involved and active in all video presentations.

- The laboratory is presented in writing and orally / practically.
- The written report consists of a **report in PDF format** consisting of a front page with the course name / course code, date, and student (name and DU-identity).
 - **Part 1:** Must be reported point by point based on the list of 'objectives' that are specified in the instructions. Under each point you give a short text description of how you perform the specific task and then you include screenshots from the code itself and attach proof/test results, showing that the code works and executes like expected.
 - See this example for guidance:

PART1

TASK: Create a directory at the root (/) of the file system for each department

`cd /`

`Sudo mkdir Engineering Sales HR`

Verifying with code and output from screen:

`ls -l`

```
sysadmin@localhost:~$ cd /
sysadmin@localhost:/$ sudo mkdir Engineering Sales HR
[sudo] password for sysadmin:
sysadmin@localhost:/$ ls -l
total 92
drwxr-xr-x  2 root root 4096 Oct  9 18:08 Engineering
drwxr-xr-x  2 root root 4096 Oct  9 18:08 HR
drwxr-xr-x  2 root root 4096 Oct  9 18:08 Sales
```

- **Part 2:** First describe in text how your script is supposed to work step by step, which elements should be included and how these should act according to the list of tasks (a – i) in the instruction. Then report screenshots from the actual implementation of the script, as well as execution/testing of all functions of the script and outcomes from these tests.
- The oral / practical part: You should **record a movie** with sound where you explain the code and show that the code works – do not forget to run all your tests and show their results in the video to verify that the script is working just fine . **Make sure your movie-recording really works before submitting.**
 - If you like, you can upload the movie to YouTube and enter the link to the movie in the lab report. **Make sure the link really works before submitting the lab.**
 - All students who participate in the group's presentation **must be involved and active during the film presentation!**

HAND IN YOUR WRITTEN REPORT AND YOUR LINK TO THE VIDEO RECORDING IN THE DESIGNATED FOLDER IN THE COURSE ROOM AT LEARN!